



Write a Python function that accepts a single temperature value as input, along with a unit of measurement specified as 'C' for Celsius or 'F' for Fahrenheit (defaulting to Celsius if not provided), and a threshold value for triggering a heat advisory.

The function should convert the temperature to the opposite unit using the standard formula $C = (F - 32) \times 5/9$ for Fahrenheit to Celsius, and $F = (C \times 9/5) + 32$ for Celsius to Fahrenheit and check whether the converted temperature is below or above the given threshold. If the temperature is below the threshold, return "Cold advisory" if it is equal to or above the threshold, return "Heat alert". ensure the accuracy of the temperature conversion and advisory message logic by writing and running validation tests using TDD.



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Write a function that accepts the name of an item, the original price of the item, and a promotional code. If the promotional code is "SAVE10", apply a 10% discount, if it's "HALFOFF", apply a 50% discount and if there's no valid code apply no discount. Return the discounted price. write tests confirming that the correctness of the function and ensure you handle potential edge cases.



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Create a function that determines if a given number is both a palindrome and a prime number. The function should return True if the number meets both conditions, otherwise False. Consider efficiency and handle edge cases.



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